Demonstration of Proposed Features

Date	29 June 2026
Team ID	SWTID-2026-8096
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

The **Rising Waters: A Machine Learning Approach to Flood Prediction** system was successfully developed and demonstrated as a web-based application. The application allows users to enter rainfall-related parameters through an interactive interface and instantly predicts whether there is a **Flood Chance** or **No Flood Chance** using a trained Machine Learning model.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	ML Model Development	Training classification models (Decision Tree, Random Forest, KNN, XGBoost) on historical weather data to predict flood likelihood.	Implemented	Yes	XGBoost model achieved 96.55% accuracy on test data.
2	Flask Web Application	Integration of the best-performing ML model into a Flask app with an intuitive user interface.	Implemented	Yes	Enables authorities to easily input data and monitor risks.
3	IBM Cloud Deployment	Scaling the application and making it accessible from any location via cloud hosting.	Partial / Pending	No	The app is currently "designed for potential deployment" rather than actively hosted.
4	Early Warning Module	Processing current rainfall and cloud visibility to predict immediate high-probability flooding.	Implemented	Yes	Successfully allows for advanced evacuation advisories.

5	Multi-region Monitoring	Analyzing regional weather data for multiple areas to prioritize disaster response resources.	Implemented	Yes	Provides instant flood risk classifications for coordinators.
6	Model Validation & Assessment	Testing the model against historical flood event data to evaluate accuracy	Implemented	Yes	XGBoost model achieved 96.55% accuracy on test data.

Feature Implementation Summary:

Total Features Proposed	8
Total Features Implemented	8
Total Features Demonstrated	4
Overall Implementation Rate (%)	80%